



TEMASEK JUNIOR COLLEGE
Preliminary Examinations 2009



BIOLOGY
Higher 2
Paper 2 Core Paper

9747/02
Thursday, 17 September
2 hours

Candidate Name: _____

Civics Group: _____ /08

READ THESE INSTRUCTIONS FIRST

Write your name and civics group on all answer papers used.
Write in dark blue or black pen.
You may use a soft pencil for any diagrams, graphs or rough working.
Do not use staples, paper clips, highlighters, glue or correction fluid.

Section A

Answer **all** questions

Section B

Answer any **one** question

At the end of the examination, fasten the answer papers securely and hand up separately.

The number of marks is given in brackets [] at the end of each question or part question.

For examiner's use	
Section A	
1	
2	
3	
4	
5	
6	
7	
Section B	
8 / 9	
TOTAL	

This document consists of **18** printed pages.

Section A

For
Examiner's
Use

Answer **all** the questions in this section.

- 1 The electron micrograph in Fig. 1.1 shows an acinar cell found in the pancreas.

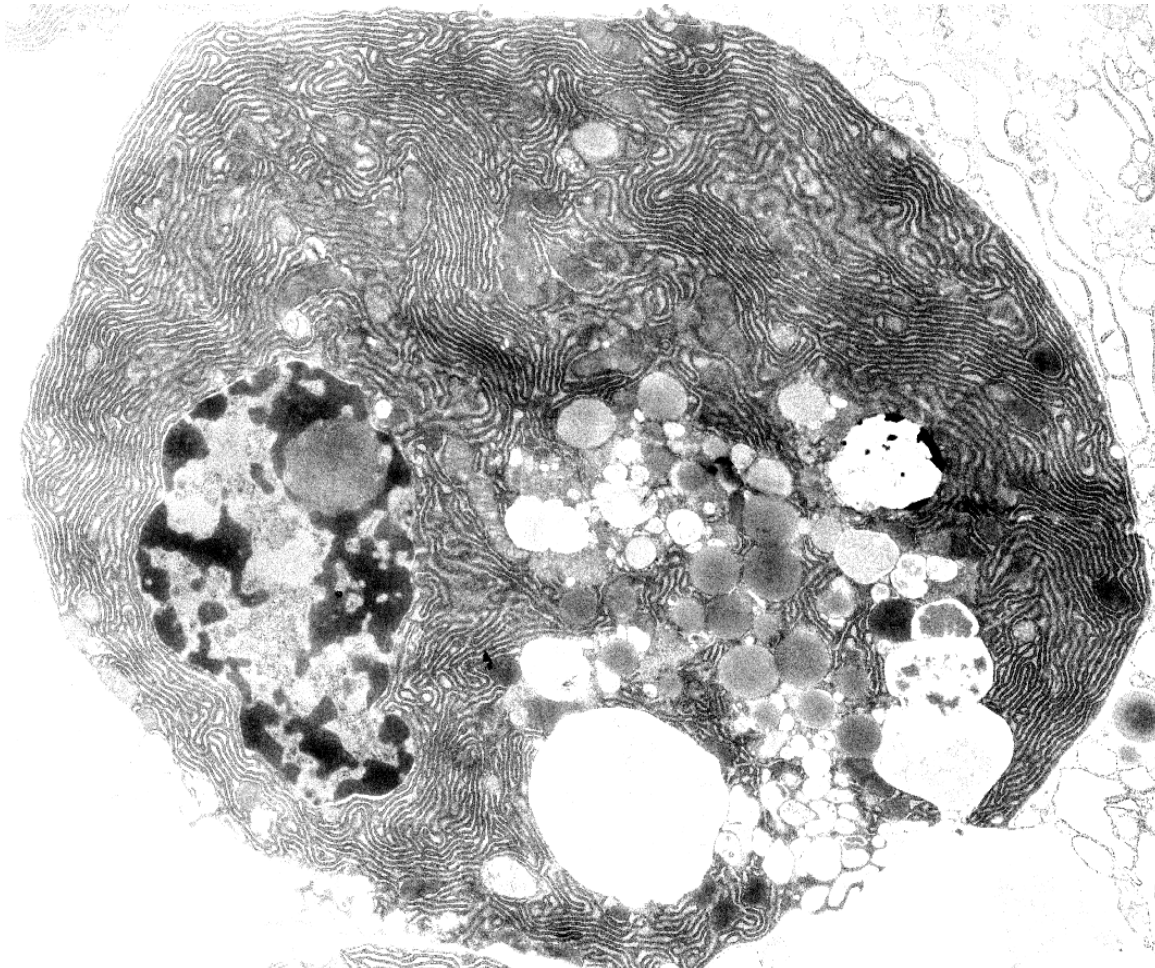


Fig. 1.1

- (a) With reference to Fig. 1.1, suggest the principal function of the pancreatic acinar cell. Give **two** reasons to explain your answer.

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.....[3]

Fig. 1.2 shows two different structures, A and B, found in an animal cell.

For
Examiner's
Use

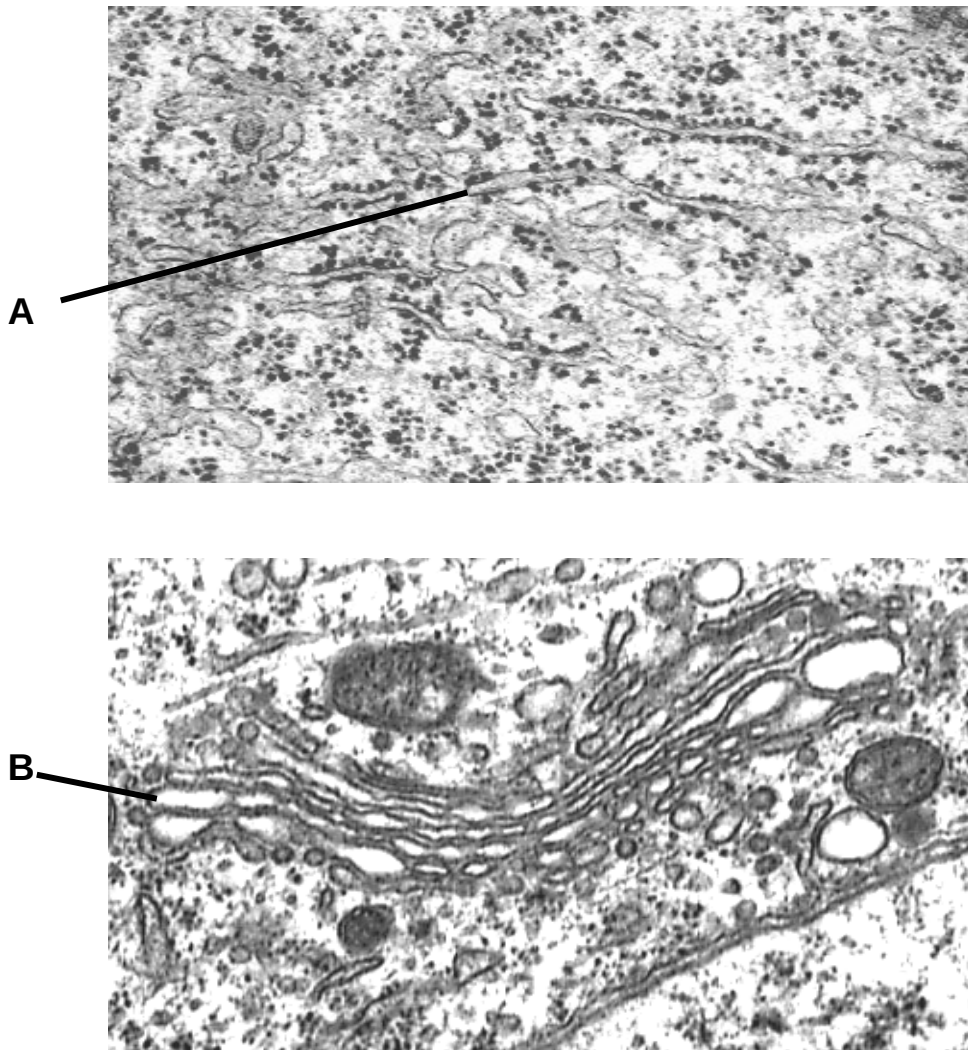


Fig. 1.2

(b) (i) Describe **three** ways in which A is similar to B.

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 [3]

(ii) The products of B are usually in the inactive form. Suggest the significance of this. [1]

.....
 [1]

- 2 Photomicrographs in Fig. 2.2 show the stages of mitosis in a bluebell (Fig. 2.1), which has a diploid number of 16.

For
Examiner's
Use



Fig. 2.1

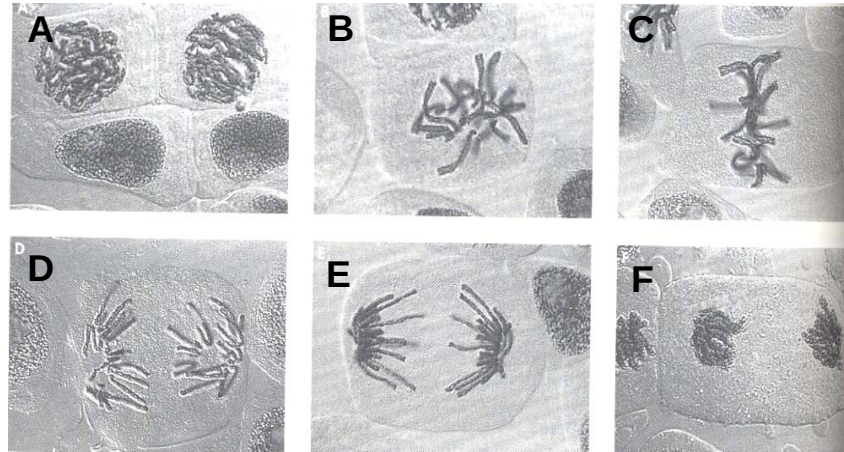


Fig. 2.2

- (a) Identify using arrow(s) and label on Fig. 2.2, which cells are undergoing interphase. Explain your answer.

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..... [2]

- (b) With reference to Fig. 2.2 C,

- (i) state the stage of division observed.

..... [1]

- (ii) draw and label cells undergoing a similar stage during formation of gametes, in the space below.

- (c) Explain the significance of synapsis during gamete formation.

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..... [3]

Fig. 2.3 shows changes in DNA content at different stages of the animal life cycle.

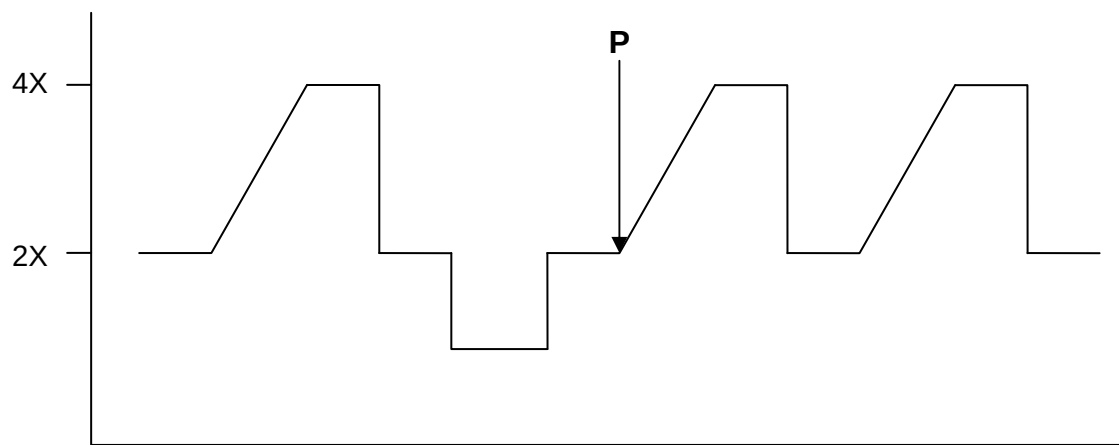


Fig. 2.3

- (d) With reference to Fig. 2.3,

- (i) identify and describe **Stage P**.

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..... [3]

- (ii) state the significance of **Stage P**.

..... [1]

[Total: 12]

[Turn over

- 3** In prokaryotes, genes are often organized into operons. Some examples of well-studied operons include the *lac* and *trp* operons, which are involved in lactose metabolism and tryptophan synthesis respectively.

(a) Explain the meaning of the term 'operon'.

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..... [3]

(b) Discuss the similarity between the control of the *trp* and *lac* operons.

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..... [2]

- (c) With the use of named examples, discuss the differences between inducible and repressible operons.

For
Examiner's
Use

Inducible operons	Repressible operons
Named example:	Named example:

[4]

A certain temperate bacteriophage is known to insert its own genome within the *trp* operon, downstream of the promoter. The genome of the bacteriophage contains excision genes. When expressed, these genes will result in the excision of the prophage from the host genome.

- (d) Explain how changes in the amount of tryptophan present in the immediate vicinity of bacteria infected by the phage might affect the numbers of free bacteriophage detected.

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..... [4]

[Total: 13] **Turn over**

- 4 Fig. 4.1 shows an electron micrograph of *Escherichia coli*.

For
Examiner's
Use

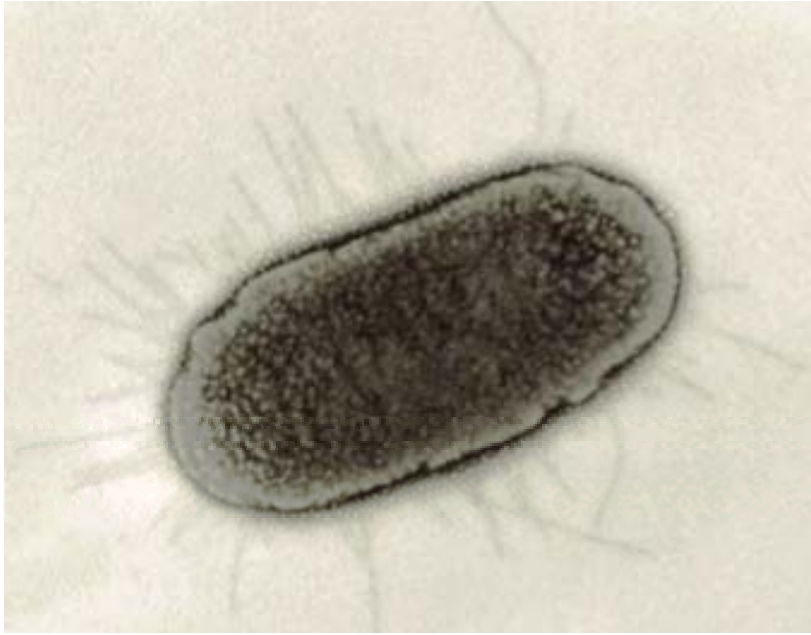


Fig. 4.1

- (a)** Describe the structural organization of DNA in a bacterial cell.

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..... [3]

Bacteria reproduce by a process called binary fission.

- (b) (i)** Explain the significance of binary fission in bacteria replication.

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..... [4]

(ii) Distinguish between binary fission and mitosis.

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..... [2]

Bacteria can reproduce at extremely rapid rates. An *E.coli* cell is capable of dividing every 20 minutes when grown under optimal conditions. However, experiments have shown that the time taken to completely replicate one chromosome is often longer than the average time taken for the cell to divide.

(c) (i) Suggest how it is possible for a bacterium to divide at a rate faster than that at which its DNA can be replicated.

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..... [2]

(ii) Given that the *E.coli* chromosome contains 4 million nucleotides, and that the rate of DNA replication occurs at 1000 nucleotides per second, calculate the time required for an entire *E.coli* chromosome to be replicated.

..... [1]

[Total: 11]

- 5** Three different feather colours have been observed in Andalusian fowl – black, white and blue. Comparison of the alleles for white feathers and black feathers showed that there was a deletion of a single base-pair in the allele coding for white feathers.

To determine the mode of inheritance of feather colour in these fowl, several types of crosses were carried out.

In the parental generation, 2 crosses were carried out:

1. Female, pure-breeding black feathered fowl x male, pure-breeding white feathered fowl
2. Male, pure-breeding black feathered fowl x female, pure-breeding white feathered fowl

All of the F_1 generation in both crosses had blue feathers.

Sibling mating was then carried out within the F_1 generation, and the following results were obtained:

Black feathered fowl: 439

White feathered fowl: 442

Blue feathered fowl: 879

- (a)** State the type of cross carried out in the parental generation.

..... [1]

- (b)** Explain why a reciprocal cross was **not** carried out within the F_1 generation.

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..... [2]

- (c)** Using suitable symbols, construct a genetic diagram to explain the results of the F_1 sibling mating.

Fireflies contain a compound called luciferin, which can be oxidised by the enzyme luciferase, to oxyluciferin. Light is emitted in the process of this reaction.

Researchers have discovered that D-cystine and 2-cyano-6-hydroxybenzothiazole undergo a condensation reaction to form luciferin. This reaction is mediated by enzyme A.

To study the genetics of light emission in fireflies, some researchers have bred mutant fireflies that have mutations that result in non-functional enzyme A and luciferase. These mutant fireflies do not emit light.

The alleles **A** and **L** code for functional enzyme A and luciferase respectively, while the alleles **a** and **l** code for non-functional enzyme A and non-functional luciferase respectively.

Fig. 5.1 shows biochemical pathway that eventually results in the emission of light in fireflies.

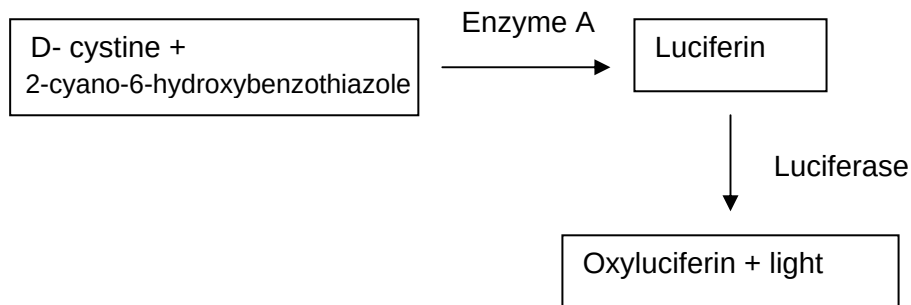


Fig. 5.1

Pure-breeding normal fireflies were crossed with pure-breeding mutant fireflies. The F_1 generation were all phenotypically normal and could emit light. The F_1 generation was then sibling mated, and the results of the sibling mating are as follows:

Light emitting fireflies:	169
Mutant fireflies:	131

(d) (i) State the term used to refer to this type of gene interaction.

..... [1]

- (ii) Explain how this type of gene interaction can result in the experimental ratios observed for the sibling mating of the F_1 generation.

*For
Examiner's
Use*

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..... [4]

[Total: 13]

- 6** The Orchid family is the largest family of flowering plants. Since the 19th century, horticulturists have produced more than 100 000 hybrids and cultivated varieties. Orchids are cosmopolitan, occurring in almost every habitat apart from deserts and glaciers. Some orchids are saprophytes and they have no need for leaves, as all their nutrition is gained from dead vegetation and animal material. They only appear above ground to flower and seed. One such species is the Ghost Orchid *Epipogium gmelini*, found only in the deepest beech woods. Another is the Bird's Nest Orchid, which at first glance, can be easily mistaken for a member of the Broomrape family Orobanchaceae, where all are parasitic and feed from the roots of other plants.

Orchid conservationists have so hard a time trying to germinate orchid seeds and grow orchids in an artificial environment. Orchid seeds are tiny, frail and delicate that contain virtually nothing except an embryo. They must enter symbiotic relationship with various fungi that provide them the necessary nutrients to germinate. Orchids have developed highly specialized pollination systems. In some extremely specialized orchids, the flower is adapted to have a colour, shape and odour which attracts male insects via mimicry of a receptive female. Pollination happens as the insect attempts to mate with flowers.

- (a)** Explain if the biological species concept is suitable to study the Orchid family.

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..... [4]

- (b)** Between the morphological and the ecological species concepts, suggest which will be more useful for research on the Bird's Nest orchid.

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..... [3]

- (c) State **one** advantage and **one** disadvantage of using an artificial classification as compared to a natural classification in taxonomy.

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..... [2]

- (d) With reference to the passage, state **two** advantages of being saprophytic.

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..... [2]

- (e) Suggest why orchid conservationists have difficulty germinating orchid seeds.

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..... [1]

[Total: 12]

7

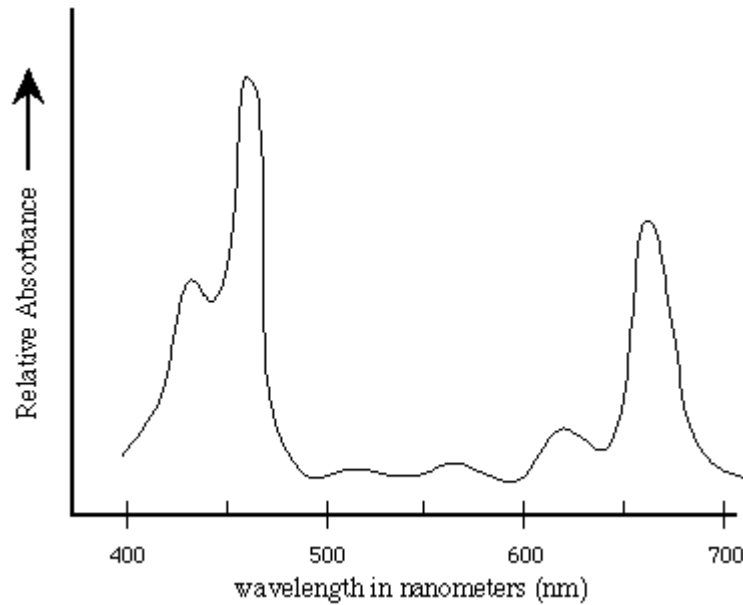


Fig. 7.1

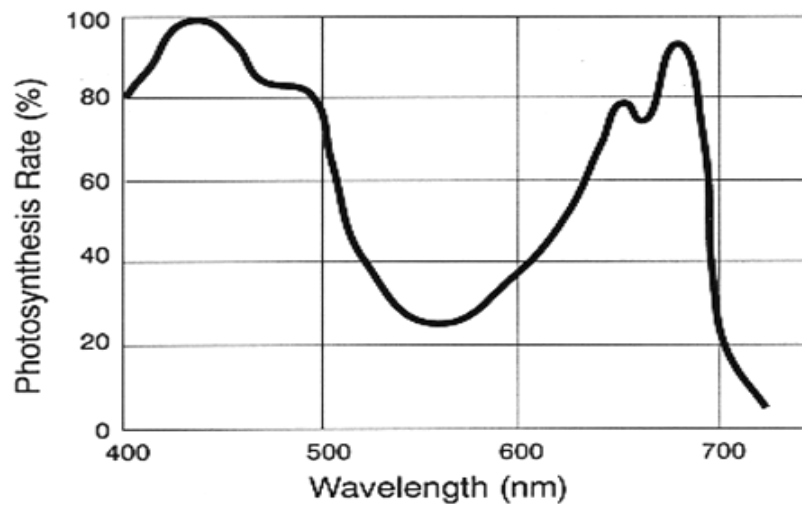


Fig. 7.2

- (a)** With reference to Fig. 7.1 and Fig. 7.2, explain the role of light harvesting complex in both photosystems I and II. [3]

.....[3]

[Turn over

- (b) Distinguish the electron transport chain in photophosphorylation from that in oxidative phosphorylation.

Photophosphorylation	Oxidative Phosphorylation
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[3]

The ATP and NADPH synthesized during photophosphorylation are utilised in the Calvin cycle for the synthesis of glucose, which is subsequently oxidised by the cell to produce ATP for its biochemical processes. Glucose may also be converted to sucrose and transported to the roots to be stored as starch.

- (c) Suggest a reason why ATP generated in the chloroplast is **not** used directly for the cell's biochemical processes.

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..... [1]

(d) (i) In the space given, draw a monomer of starch.

[1]

(ii) Distinguish the molecule you have drawn in **(d)(i)** from a monomer of cellulose.

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 [1]

In animals, triglycerides serve as long term storage of energy, rather than carbohydrates.

(e) Suggest why triglycerides are better storage molecules than carbohydrates in animals. [3]

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 [3]

[Total: 12]

[Turn over]

Section B

Answer **one** question.

Write your answers on the separate answer paper provided.

Your answers should be illustrated by large, clearly labelled diagrams, where appropriate.

Your answers must be in continuous prose, where appropriate.

Your answers must be set out in sections **(a)**, **(b)** etc., as indicated in the question.

- 8 (a)** Describe how eukaryotic cells can regulate gene expression at the translational and post-translational levels. [4]
- (b)** Describe how an action potential is propagated along an unmyelinated axon. [6]
- (c)** Discuss, with examples, how gene and chromosomal mutations may eventually lead to cancer. [10]
- 9 (a)** Explain how the neutral theory of molecular evolution appears to be an argument against Darwin's theory of natural selection. [7]
- (b)** Using a named example, explain why a population is the smallest unit that can evolve. [5]
- (c)** Describe the homeostatic regulation of blood glucagon level. [8]